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(54) **IN-FIELD EMERGENCY BLOOD TRANSFUSION SYSTEM AND METHOD**

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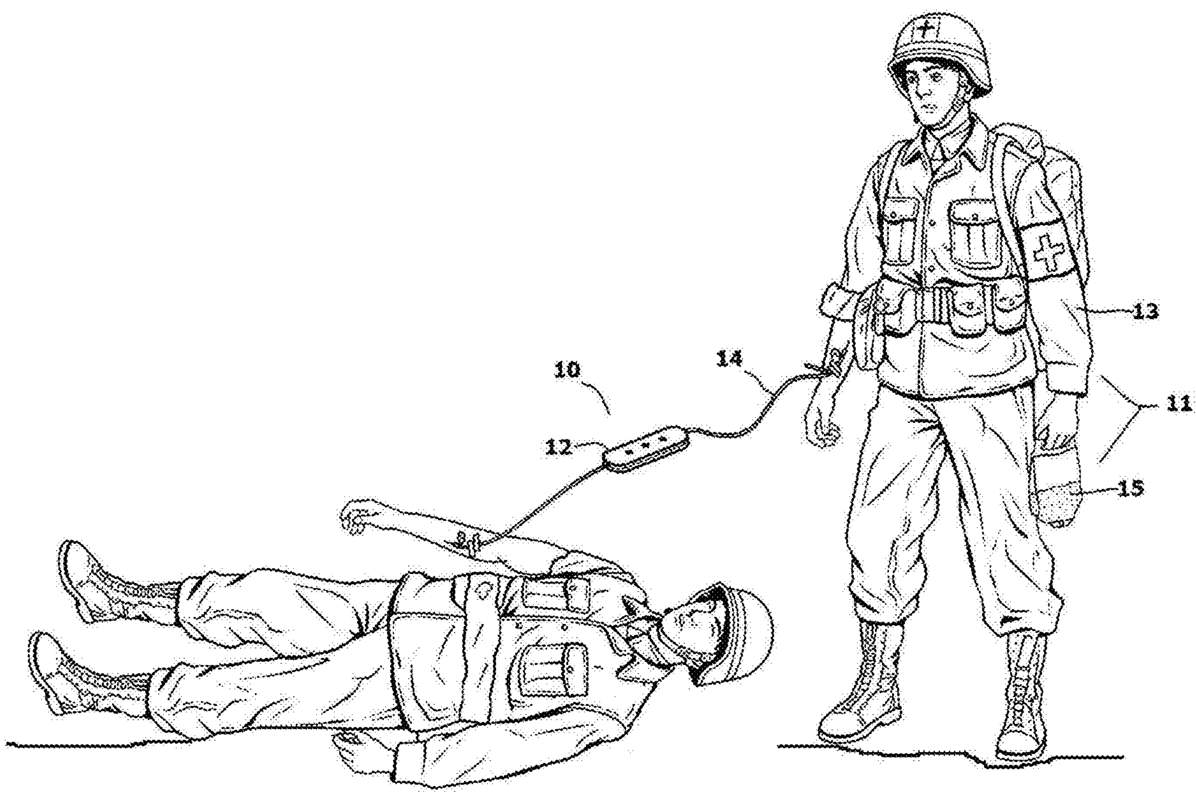
(57) **ABSTRACT**

A system and method of transfusing blood from a donor source to a patient. To facilitate the transfusion, a transfer tube is provided. One end of the transfer tube is connected to the donor source and the opposite end is intravenously connected to the patient. A pump assembly is provided that contains a pump and a controller. The transfer tube is engaged with the pump assembly wherein the pump, when activated, acts upon the transfer tube to move blood. The controller monitors the blood volume moved and automatically stops the pump once a predetermined volume of blood has been transferred. If the donor source is a person, the predetermined volume is between 400 milliliters and 450 milliliters. The flow rate of the pump is preferably 100 milliliters per minute. This transfusion rate can be increased by synchronizing the pump to the heart rhythm of the patient.



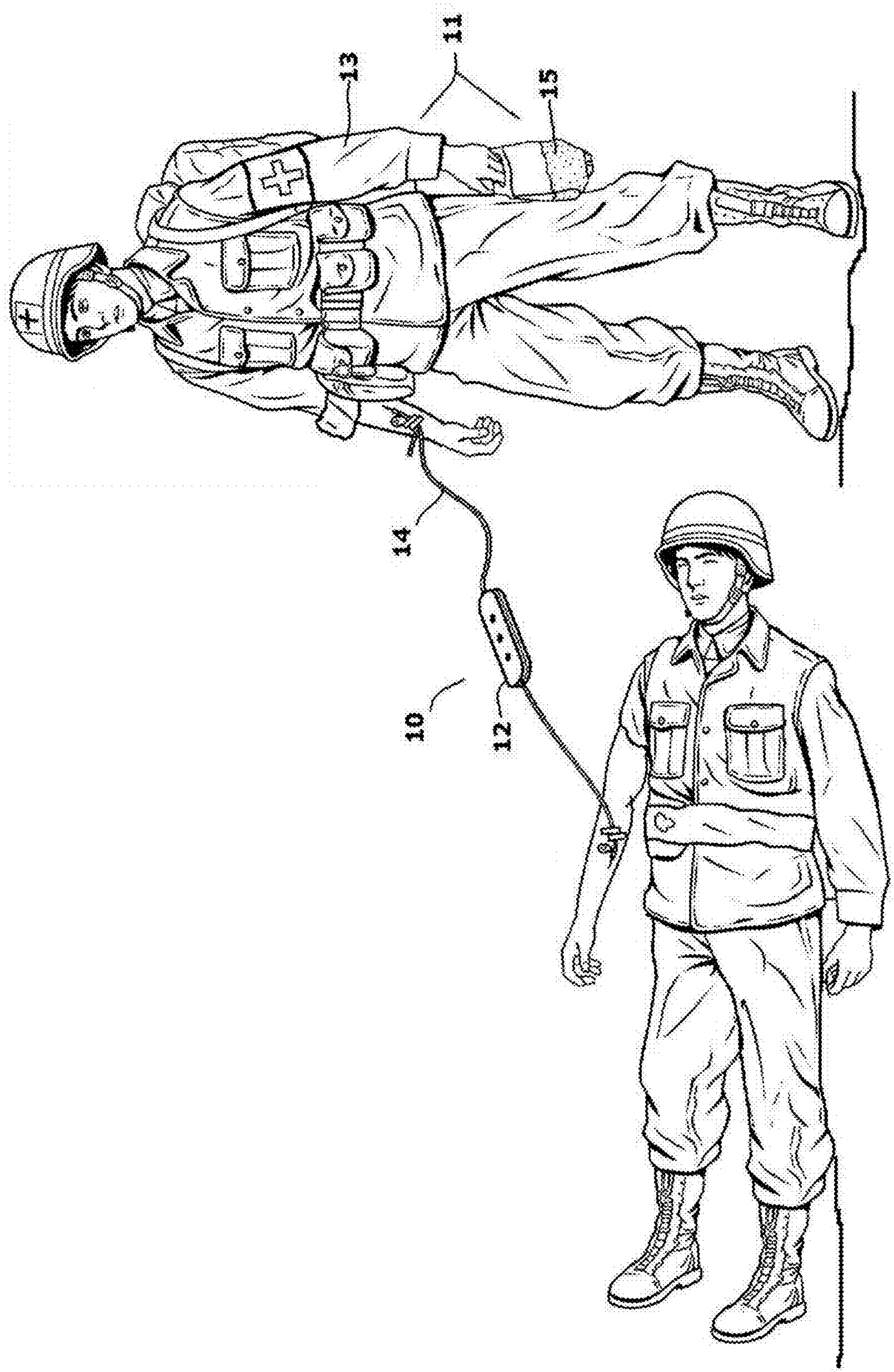


FIG. 1

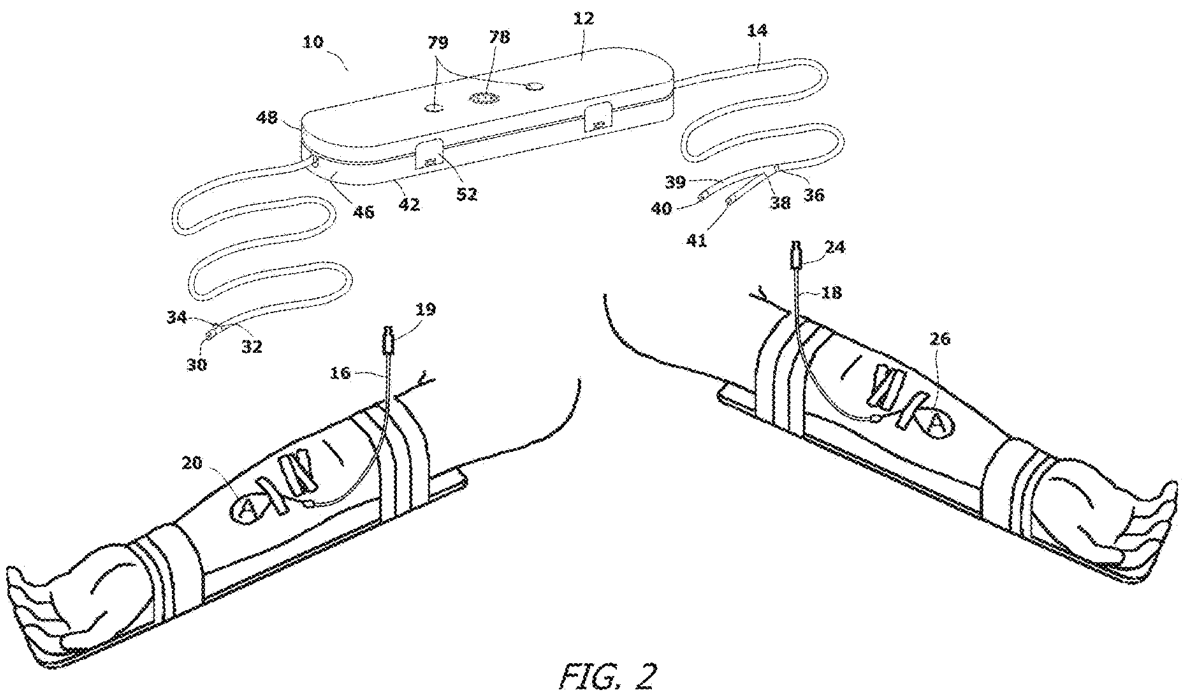


FIG. 2

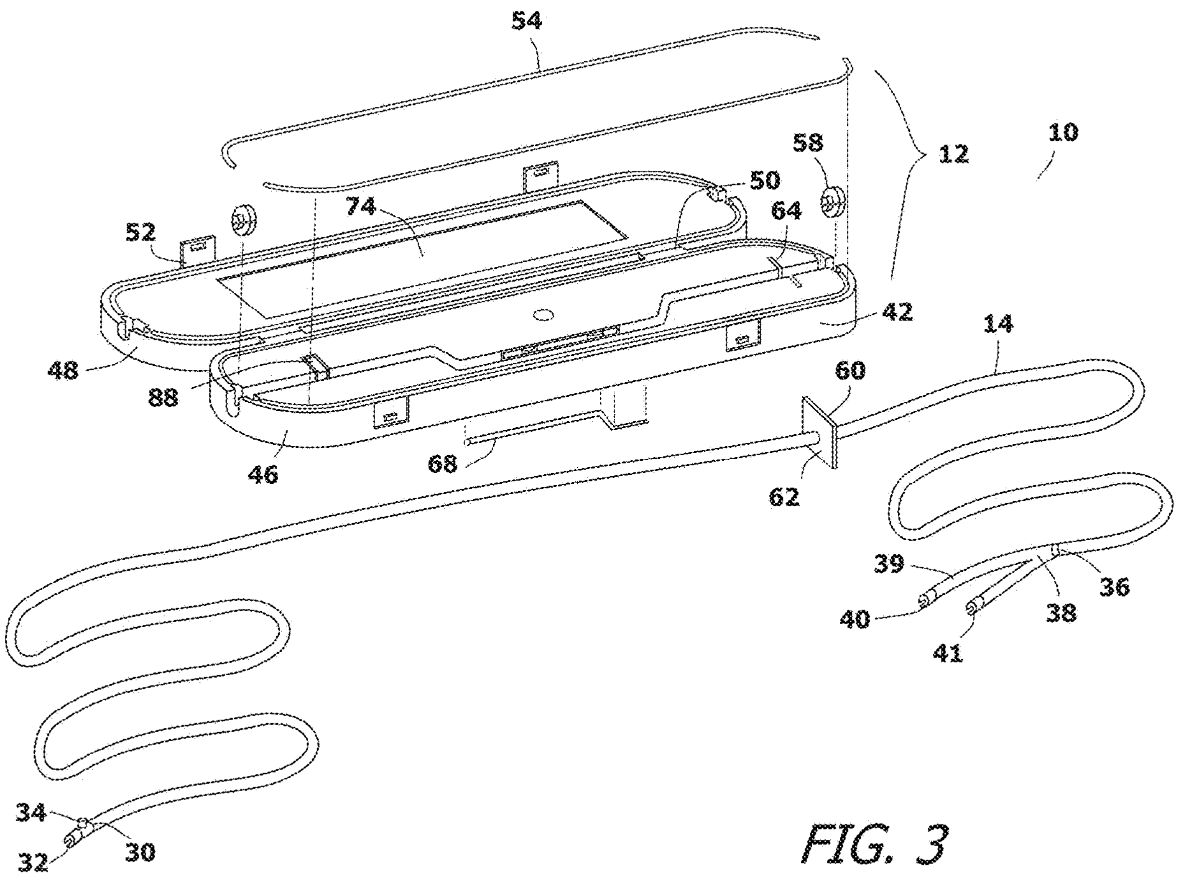


FIG. 3

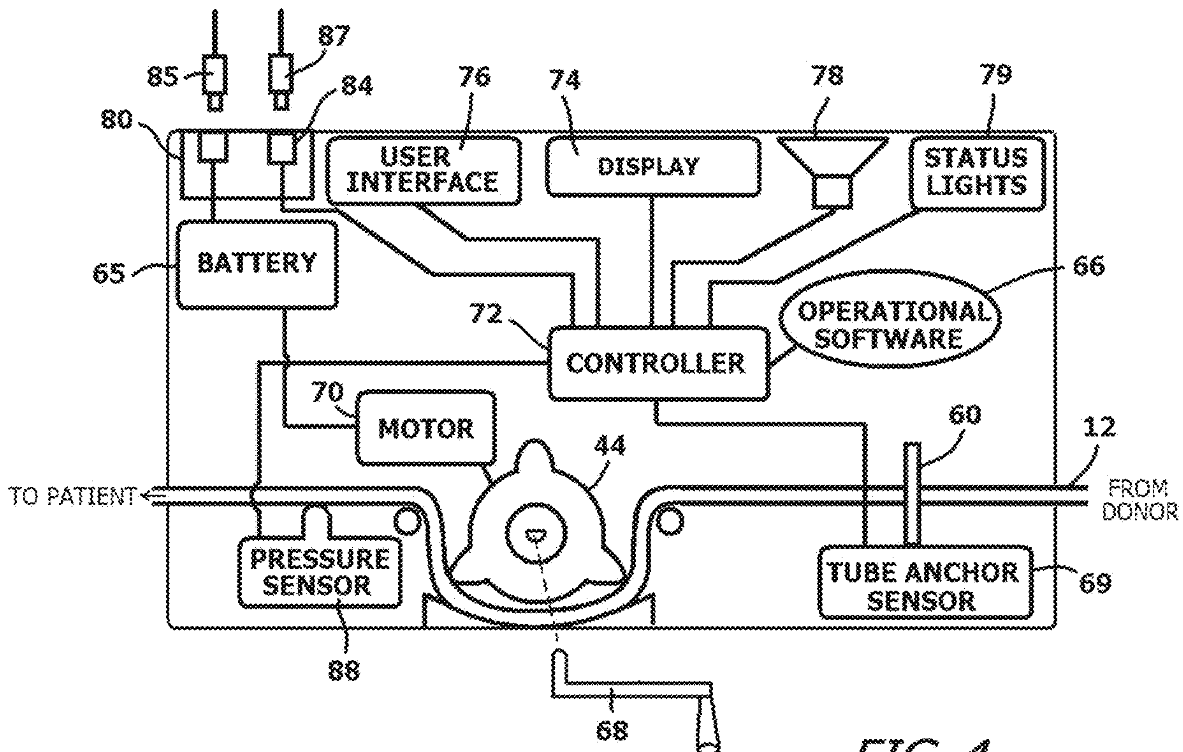

85
87
84
80
76
74
78
79
USER INTERFACE
DISPLAY
STATUS LIGHTS
65
BATTERY
66
OPERATIONAL SOFTWARE
72
CONTROLLER
70
MOTOR
44
60
12
TO PATIENT
FROM DONOR
PRESSURE SENSOR
TUBE ANCHOR SENSOR
69
88
68

FIG. 4

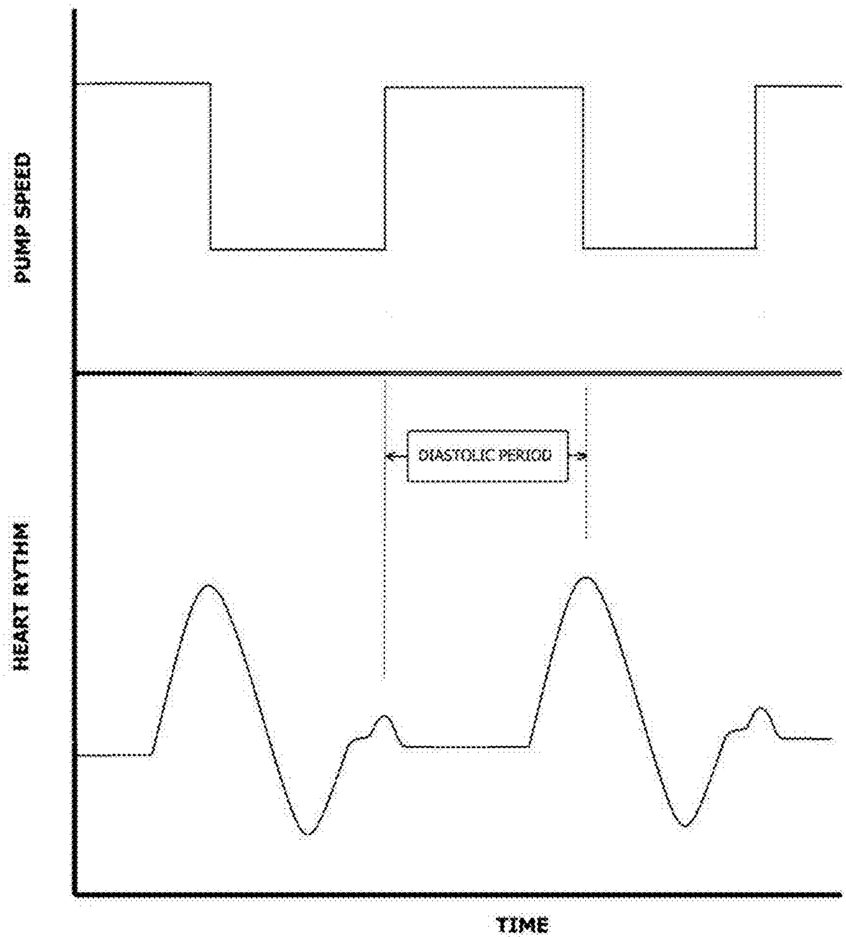


FIG. 5

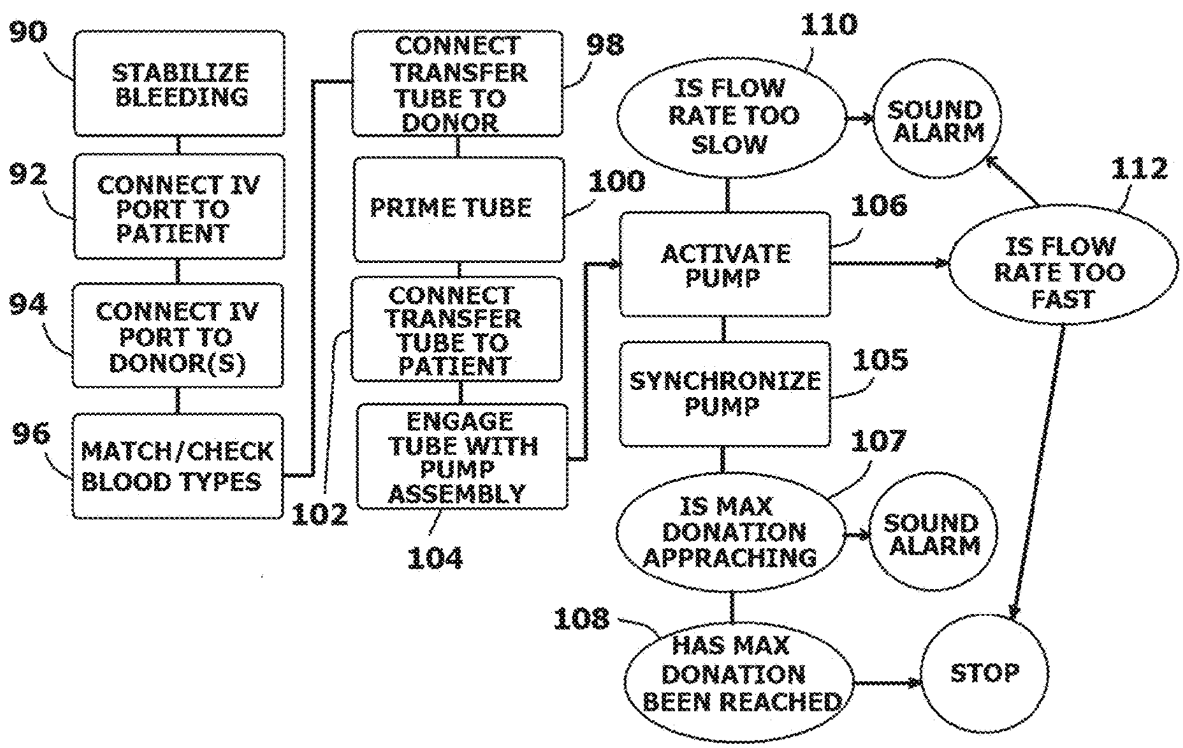

90
STABILIZE BLEEDING
92
CONNECT IV PORT TO PATIENT
94
CONNECT IV PORT TO DONOR(S)
96
MATCH/CHECK BLOOD TYPES
CONNECT TRANSFER TUBE TO DONOR
98
PRIME TUBE
100
CONNECT TRANSFER TUBE TO PATIENT
102
ENGAGE TUBE WITH PUMP ASSEMBLY
104
110
IS FLOW RATE TOO SLOW
SOUND ALARM
ACTIVATE PUMP
106
112
IS FLOW RATE TOO FAST
SYNCHRONIZE PUMP
105
107
IS MAX DONATION APPRACHING
SOUND ALARM
108
HAS MAX DONATION BEEN REACHED
STOP

FIG. 6

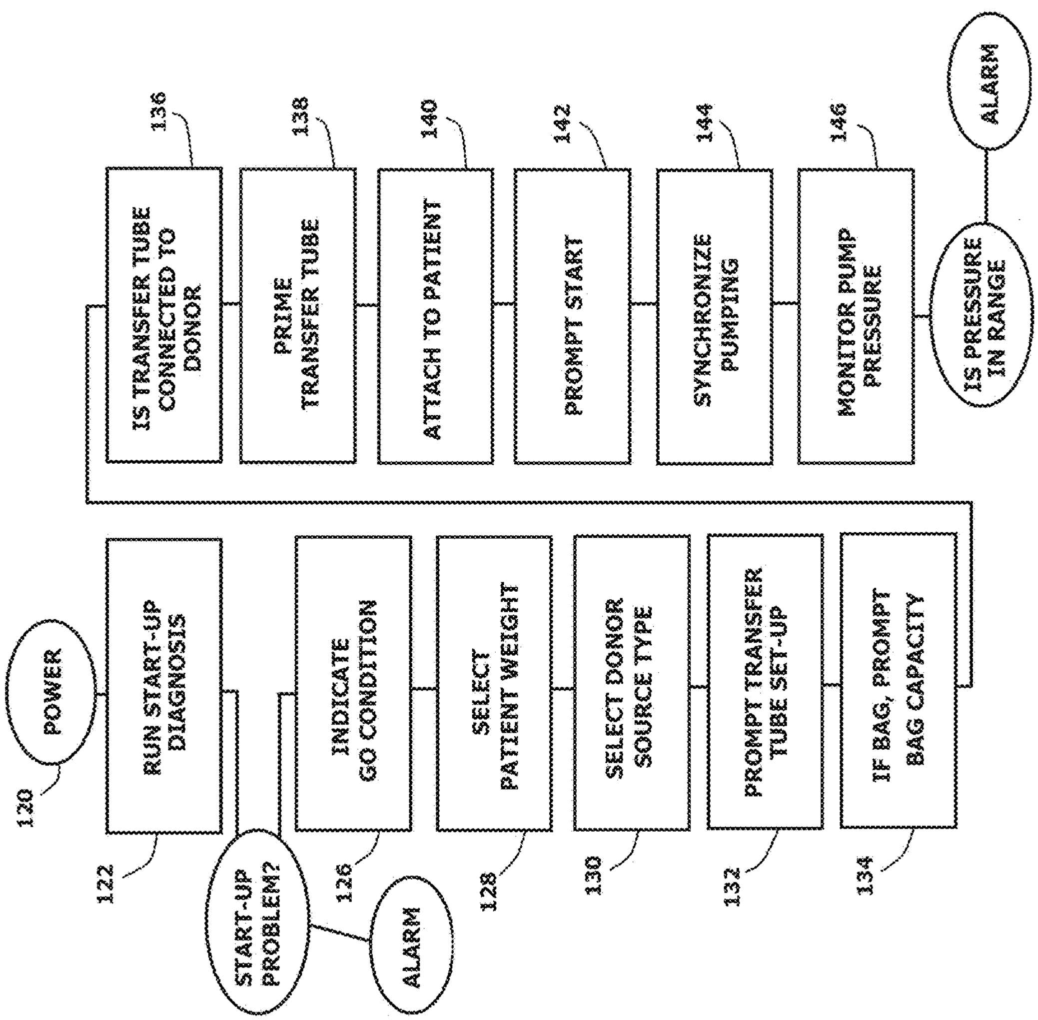

POWER
120
RUN START-UP DIAGNOSIS
122
START-UP PROBLEM?
ALARM
INDICATE GO CONDITION
126
SELECT PATIENT WEIGHT
128
SELECT DONOR SOURCE TYPE
130
PROMPT TRANSFER TUBE SET-UP
132
IF BAG, PROMPT BAG CAPACITY
134
IS TRANSFER TUBE CONNECTED TO DONOR
136
PRIME TRANSFER TUBE
138
ATTACH TO PATIENT
140
PROMPT START
142
SYNCHRONIZE PUMPING
144
MONITOR PUMP PRESSURE
146
IS PRESSURE IN RANGE
ALARM

FIG. 7

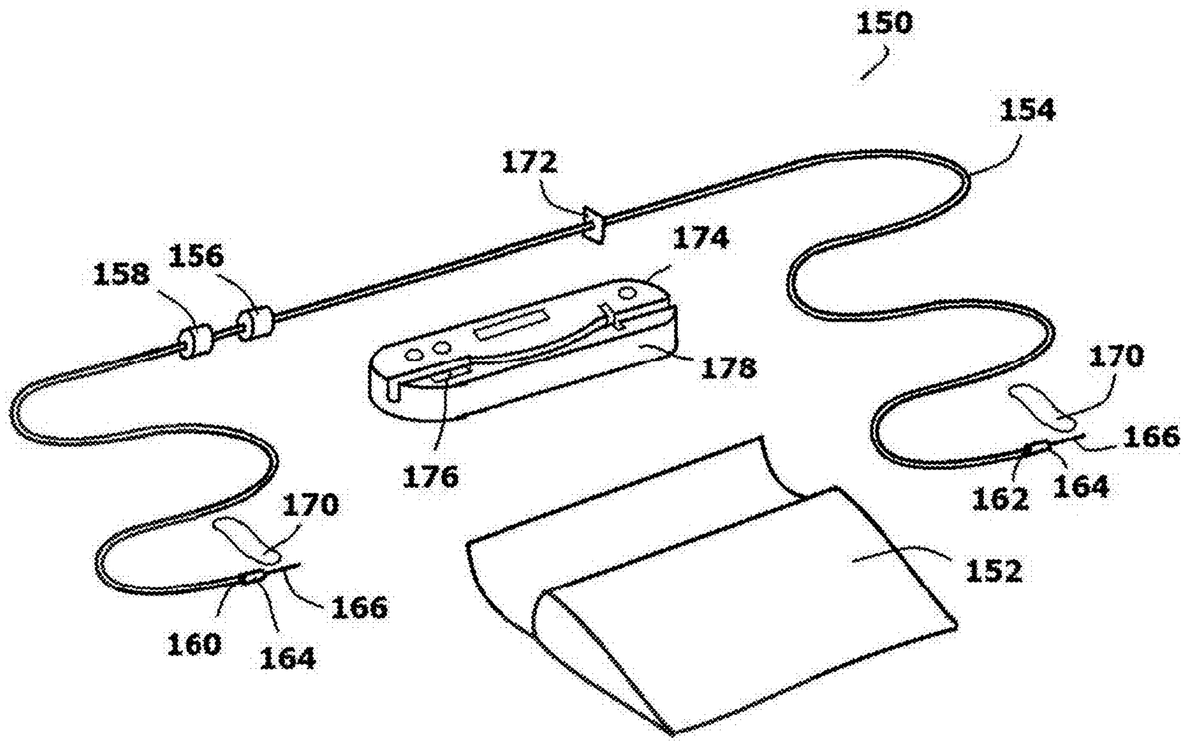

150
154
172
174
158
156
178
170
166
176
162
164
170
152
166
160
164

FIG. 8

IN-FIELD EMERGENCY BLOOD TRANSFUSION SYSTEM AND METHOD

RELATED APPLICATION

[0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/560,154, filed Mar. 1, 2024.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] In general, the present invention relates to blood transfusion systems that are used in emergency situations, such as during natural disasters or on the battlefield. More particularly, the present invention relates to transfusion systems with pumps and tubes that can transfer blood from a donor source to a person in need in a highly expedited manner.

2. Prior Art Description

[0003] The average adult male human body holds about 2000 milliliters of blood. If a person loses one-fifth of their blood they are likely to go into hemorrhagic shock and lose consciousness. If a person loses one fourth of their blood, they are likely to die from exsanguination. Likewise, regardless of blood loss, if a person's blood pressure drops below 70/30 they are in danger of dying. In addition, even if the loss of blood does not cause death, the longer the state of hemorrhagic shock lasts, the more damage is caused to the body. Prolonged hemorrhagic shock can cause heart damage, brain damage, tissue loss and gangrene.

[0004] Blood loss to the point of exsanguination is typically caused from physical trauma, such as a gunshot wound, a shrapnel wound, or an impalement. The loss of blood is directly related to the location and size of the wound. In many instances, the wound can be bound to a degree where blood loss is manageable. However, if the patient has already lost too much blood, the blood must be replaced before the body passes into shock or passes from shock to death.

[0005] In cases of rapid blood loss, patients are typically treated using a blood transfusion. An intravenous line is inserted into the patient and a stored bag of blood or plasma is attached to the line. The blood or plasma then flows into the patient using the force of gravity. The typical flow of blood into the body using this system is typically around 70 ml per minute. This rate can be increased by using an intravenous line pump. However, even with such a pump, the maximum flow rate is around 80 ml per hour. Most intravenous line pumps are designed to introduce intravenous fluids, such as medicated saline solutions into a patient. Intravenous line pumps are designed to supply controlled volumes that are often measured in drops per minute, which translates to milliliters per hour. Intravenous line pumps are therefore designed for small flow rates and are not intended to maximize flow from a fluid source to the patient. Such prior art intravenous line pumps are exemplified by U.S. Pat. No. 9,677,555 to Kamen et al., U.S. Pat. No. 10,342,921 to Barnes et al., U.S. Pat. No. 5,399,166 to Laing and U.S. Patent Application Publication No. 2004/0064097 to Peterson.

[0006] Although such prior art intravenous line pumps can be considered transportable, they are still intended for use in a hospital or other stable environment, such as a field hospital, where there are clean conditions and access to external power. Such prior art transfusion pumps are also designed for use on stabilized patents where the flow rate through the intravenous line can be kept under 300 ml per hour. Such prior art pump systems tend to contain drip chambers that only work when vertically oriented and therefore are poorly suited for battlefield use. In addition, such prior art systems cannot operate in battlefield conditions where there is no available external power, and the unit may be soaked with rain, snow, seawater, blood, and battlefield debris.

[0007] In many circumstances, a patient may need blood immediately and no stored blood or plasma is available. This is often the case on the battlefield. The transfusion is often conducted on an active battlefield while cramped behind limited cover. In such a circumstance, a medic may have to resort to a direct donor-to-patient blood transfusion, which is often referred to as an in-field blood transfusion. In an in-field blood transfusion, blood is directly transferred from a donor to a patient. The donors must remain available and stationary for the period of transfusion, which can be as long as a half hour per donor. This is highly impractical and often impossible for soldiers on a battlefield.

[0008] A need therefore exists for an improved system and methodology of performing an in-field blood transfusion, wherein the time required to transfer blood from a donor to a patient can be greatly reduced. A need also exists for an improved system and methodology of performing an in-field blood transfusion that can operate in any orientation and robust enough to function without external power in a very wet and contaminated environment. This need is met by the present invention as described below.

SUMMARY OF THE INVENTION

[0009] The present invention is a system and method of rapidly transfusing blood directly from a donor source to a patient. The donor source can be a living person or a prefilled bag of blood or blood plasma. To facilitate the transfusion, a transfer tube is provided. One end of the transfer tube is connected to the donor source. The opposite end of the transfer tube is intravenously connected to the patient.

[0010] A pump assembly is provided that contains a pump, at least one battery for powering the pump, and a controller for controlling the pump. The pump, battery and controller are encased in a common waterproof housing. The transfer tube is engaged with the pump assembly wherein the pump, when activated, acts upon the transfer tube to move blood from the donor source to the patient.

[0011] The controller monitors blood volume moved by the pump and automatically stops the pump once a predetermined volume of blood has been transferred. If the donor source is a person, the predetermined volume is between 400 milliliters and 450 milliliters. The flow rate of the pump is preferably 100 milliliters per minute. As such, a transfusion from a human donor to a patient should take no longer than 4.5 minutes. This transfusion rate can be increased by synchronizing the pump to the heart rhythm of the patient. The rapid rate of transfusion represents a significant advancement in the art that can save many patients from passing into hemorrhagic shock or passing from hemorrhagic shock to death.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] For a better understanding of the present invention, reference is made to the following description of exemplary embodiments thereof, considered in conjunction with the accompanying drawings, in which:

[0013] FIG. **1** shows an image of two soldiers using an exemplary embodiment of the present invention blood transfusion system in the field;

[0014] FIG. **2** shows an overview of an exemplary embodiment of the present invention blood transfusion system shown in conjunction with a donor and a patient;

[0015] FIG. **3** is a schematic of the major components contained within the pump assembly of the present invention;

[0016] FIG. **4** us a block diagram schematic showing the primary operational components contained within the pump assembly;

[0017] FIG. **5** is a graph that shows the synchronization of the pump to the diastolic cycle of the patient's heartbeat;

[0018] FIG. **6** is a block diagram that outlines the methodology of using the present invention blood transfusion system;

[0019] FIG. **7** is a block diagram that outlines the operational methodology of the blood transfusion system; and

[0020] FIG. **8** shows an overview of an alternate exemplary embodiment of the present invention blood transfusion system.

DETAILED DESCRIPTION OF THE DRAWINGS

[0021] Although the present invention system and methodology can be embodied in many ways, only a few exemplary embodiments are illustrated. The exemplary embodiments are being shown for the purposes of explanation and description. The exemplary embodiments are selected in order to set forth some of the best modes contemplated for the invention. The illustrated embodiments, however, are merely exemplary and should not be considered limitations when interpreting the scope of the appended claims.

[0022] Referring to FIG. **1**, FIG. **2**, and FIG. **3**, a first embodiment of an emergency blood transfusion system **10** is shown. The emergency blood transfusion system **10** contains a portable pump assembly **12**, a transfer tube **14**, and two intravenous ports **16**, **18**. These elements are transferred together in a unit package that is carried by a field medic or other medically trained personnel. The purpose of the emergency blood transfusion system **10** is to transfer a monitored volume of blood directly from a donor source to a patient in the shortest amount of time possible. The donor source **11** can be a person **13** or a prefilled bag **15** of blood or plasma.

[0023] To utilize the emergency blood transfusion system **10**, the first intravenous port **16** is inserted into a patient who is in need of blood. The first intravenous port **16** is firmly taped in place to prevent any inadvertent disengagement caused by the pressures generated by the emergency blood transfer system **10**. The first intravenous port **16** is a standard IV port that has a first tube connector **19**. A patient blood-type identifying graphic **20** is provided on the first intravenous port **16** for a purpose later explained. The blood-type identifying graphic **20** contains the letter(s) and polarity of the blood type along with a color scheme associated with that blood type. For example, many organizations including the American Red Cross® use pink to color code A+ blood and light blue to color code O-blood.

[0024] Blood or plasma is drawn from the donor source **11**. The donor source **11** can be a person **13** or a prefilled bag **15** of blood or plasma, as indicated in FIG. **1**. If no preexisting bag of blood or plasma is available, the second intravenous port **18** is connected to a person **13**. If multiple donors are available, each can be fitted with such an intravenous port. Each second intravenous port **18** preferably consists of a short tube segment **22** that terminates with a second tube connector **24**. A donor blood-type identifying graphic **26** is provided for a purpose later explained. The identifying graphic **26** contains the letter(s) and polarity of the donor's blood type along with a color scheme associated with that blood type.

[0025] A transfer tube **14** is provided. The transfer tube **14** is a medical grade silicone tube or PTFE tube with a preferred inner diameter of between 1.6 mm and 5.0 mm. The transfer tube **14** is capable of interacting with the portable pump assembly **12** so that the portable pump assembly **12** can move fluid through the transfer tube **14**. The transfer tube **14** can have any length between one and three meters. The transfer tube **14** has a first end **30** that terminates with a first port connector **32**. The first port connector **32** is configured to selectively interconnect to the first tube connector **19** on the patient's first intravenous port **16**. The first port connector **32** also has a venting valve **34** that enables the transfer tube **14** to be selectively vented. This enables air to exit the transfer tube **14** at the venting valve **34** during a priming procedure.

[0026] The opposite second end **36** of the transfer tube **14** preferably has a Y-junction **38**. Each arm **39**, **41** of the Y-junction terminates with a second port connector **40** that can be attached to the second tube connector **24** on the intravenous port **18** of the donor. The Y-junction **38** enables the transfer tube **14** to be connected both to a first donor and/or to a bag of blood or plasma.

[0027] Alternatively, the Y-junction **38** enables the transfer tube **14** to be connected to a first donor and then to a second donor just prior to the disconnection of the first donor. This enables a constant flow of blood from subsequent donors without having to pause the flow of blood and without needing to reprime the transfer tube **14**. It also allows a bag of blood or plasma to be substituted for a donor and vice versa.

[0028] The flow of blood through the transfer tube **14** is not controlled by gravity and is unaffected by gravity or orientation. Rather, the portable pump assembly **12** is used. As is shown best in FIG. **2** and FIG. **3**, the portable pump assembly **12** has a containment housing **42**. The containment housing **42** is designed for battlefield use and provides a hermetic barrier around the electronics and other working components in accordance with the casing standard outlined in military standard specification MIL C-4150J. The containment housing **42** is also crushproof by military standards. That is, the containment housing **42** is capable or maintaining integrity under 100 psi of pressure. Thus, the containment housing **42** can be stepped on or even run over by a vehicle without damage.

[0029] Referring to FIG. **2** and FIG. **3** in conjunction with FIG. **4**, it will be understood that the containment housing **42** is used to hold an electromechanical pump **44**, as well as the power source, electronics, and user interface for operating the electromechanical pump **44**. In the shown embodiment, the containment housing **42** is shown having a clamshell design wherein a first housing section **46** closes over a second housing section **48** using a hinged connection **50**. The clamshell design minimizes the overall size of the portable pump assembly **12**. In a closed configuration, the first housing section **46** can be locked to the second housing section **48** using traditional latches **52**. An O-ring seal **54** is provided between the first housing section **46** and the second housing section **48**. The O-ring seal **54** prevents moisture and/or debris from entering the containment housing **42** when the containment housing **42** is locked into its closed configuration.

[0030] A groove **56** is formed in the first housing section **46**. When the containment housing **42** is in an open configuration, the full length of the groove **56** is readily accessible. Plug elements **58** may be provided at opposite ends of the groove **56**. The plug elements **58** block the groove **56** and prevent moisture and debris from entering the groove **56** when the containment housing **42** is in its closed configuration. When the containment housing **42** is in its open configuration, the plug elements **58** can be manually removed to fully open the groove **56**. Secondary seals can be provided that form a seal around the transfer tube **14** once the transfer tube **14** is set into the groove **56**.

[0031] The groove **56** has a width that is slightly larger than the diameter of the transfer tube **14**. Accordingly, the groove **56** is sized to receive a section of the transfer tube **14** and hold the transfer tube **14** in position where it can be acted upon by the electromechanical pump **44**. A tube anchor **60** is used to hold the transfer tube **14** in a fixed position relative to the electromechanical pump **44**. The tube anchor **60** is a flange **62** that is affixed to the transfer tube **14** at a specific point. A flange slot **64** is formed in the containment housing **42** along the groove **56** that receives and retains the tube anchor **60**. The engagement of the tube anchor **60** into the flange slot **64** prevents the transfer tube **14** from moving relative to the containment housing **42** as the electromechanical pump **44** applies forces to the transfer tube **14**. A sensor **69** is present in the flange slot **64** that can detect the presence of the flange slot **64** and provide a signal.

[0032] In FIG. **4**, the electromechanical pump **44** is shown as a peristaltic pump. Such a pump is preferred. However, other pumps, such as lobe pumps or gear pumps can also be used provided they are modified to move fluid within a flexible tube without directly contacting the fluid in the tube. The electromechanical pump **44** is used to move blood and/or plasma through the transfer tube **14**. The electromechanical pump **44** can engage the transfer tube **14** anywhere that the tube anchor **60** is provided. The electromechanical pump **44** is primarily powered by a battery **65** that is within the containment housing **42**. The battery **65** is preferably a rechargeable battery, such as the ABC batteries utilized by the U.S. Military. The rate that the electromechanical pump **44** moves blood/plasma through the transfer tube **14** is controlled by operational software **66**. The operational software **66** can also monitor the operational rate of the electromechanical pump **44** over time to calculate the volume of blood/plasma pumped during that time. The volume of blood/plasma pumped is stored by the operational software **66** for consideration by medical personnel attending to the patient. It is understood that during emergency use, the batteries **65** of the portable pump assembly **44** may drain rapidly from constant use. Replacement batteries or the ability to recharge the batteries may not exist. As such, the electromechanical pump **44** preferably has the ability to receive a manual crank **68** so that the electromechanical pump **44** can be manually turned when required.

[0033] The portable pump assembly **12** is selectively engaged with the transfer tube **14**. The electromechanical pump **44** has a motor **70** that normally powers the electromechanical pump **44**. The motor **70** is connected to a controller **72**. The controller **72** operates the motor **70** and uses the operational software **66** to monitor the rate at which the electromechanical pump **44** is turning and the number of times that the peristaltic pump turns. As such, the controller **72** can calculate the flow rate of blood/plasma through the transfer tube **14** and/or the volume of blood/plasma that has been transferred. These values can be viewed on a display **74** having a user interface **76**. The display **74** is capable of visually displaying data, instructions, and warnings. In addition to the display **74**, a speaker **78** and status light **79** are provided. The speaker **78** can audibly broadcast data, instructions, and alarms. The status lights **79** preferably contain a green light and a red light to quickly display the operational status of the system **10**.

[0034] The pressure in the transfer tube **14** between the electromechanial pump **44** and the patient is monitored by the controller **72**. A pressure sensor **88** is provided that presses against the transfer tube **14** within the groove **56**. The pressure sensor **88** is biased against the transfer tube **14**. As the pressure in the transfer tube **14** changes, the resistance to the bias changes and the pressure within the transfer tube **14** can be determined without physically contacting the blood or plasma flowing through the transfer tube **14**.

[0035] Optionally, the portable pump assembly **12** can have an input bay **80** that is protected by a waterproof closure **82**. In the input bay **80** are input ports **84**, **85** that lead to the controller **72** and/or battery **65**. The input port **85** for the battery **65** is a power lead that can engage a power cable to recharge the battery **65** or provide power in place of the battery **65**. A least one data input **84** is provided that leads to the controller **72**. The data input **84** is used to connect the controller **72** to an outside computer for program updates, data downloads, and diagnostic interrogations. The data input **84** can also be used to connect the controller **72** to an auxiliary patient sensor **86**, such as an auxiliary patient sensor **87** that is connected to the patient in need of blood. If the controller **72** can read pulse data from the auxiliary patient sensor **87**, then the operational software **66** can determine the pulse rate of the patient. The controller **72** can also use the auxiliary patient sensor **87** to detect if a pulse stops and sound an alarm.

[0036] Referring to FIG. **5** in conjunction with FIG. **4**, it will be understood that if the controller **72** is connected to an auxiliary patient sensor **87**, the operational software **66** can determine the timing of the systolic and diastolic cycle of the heart's pumping rhythm. During the systolic cycle, the heart contracts and blood pressure momentarily increases to its maximum. During the diastolic cycle, the heart expands and fills with blood, therein causing the blood pressure to momentarily decrease to its minimum. The operational software **66** operates the electromechanical pump **44** to be in rhythm with the patient's heart. The controller **72** can pulse the electromechanical pump **44** so that it pumps blood/plasma more rapidly during the diastolic cycle of the heart and less rapidly during the systolic cycle of the heart. By synchronizing the electromechanical pump **44** to the natural rhythm of the heart, the rate of blood/plasma flow into the patient can be increased by over 25 percent as compared to a constant flow pump.

[0037] Referring to FIG. **6**, in conjunction with FIG. **2** and FIG. **3**, it can be seen that to utilize the present invention transfusion system **10**, a patient's bleeding rate is stabilized and a first intravenous port **16** is connected to the patient. See Block **90** and Block **92**. A separate second intravenous port **18** is attached to a donor. See Block **94**. The first intravenous port **16** and the second intravenous port **18** contain identifying graphics **20**, **26**. The identifying graphics **20**, **26** can be color codes, number codes or other indicators of the person's blood type. In this manner, if there are many patients and many donors, a medic in an emergency situation can match the identifying graphics **20**, **26** and quickly identify what donors are proper to donate to a specific patient. See Block **96**.

[0038] Once a donor is matched to a patient, a transfer tube **14** can be connected to the donor. The transfer tube **14** is vented to allow the transfer tube **14** to fill with blood from the donor. Once primed with blood, the transfer tube **14** is connected to the first intravenous port **16** of the patient. See Block **98**, Block **100**, and Block **102**. The transfer tube **14** is then engaged with the portable pump assembly **12**. See Block **104**. The portable pump assembly **12** is then activated. See Block **106**. Once activated, the portable pump assembly **12** can draw blood from a donor at any rate that can be sustained by the donor. The preferred operating rate for an average adult male donor is 100 ml/min. As has been mentioned, the flow rate can be optionally varied so that the flow rate produced by the electromechanical pump **44** is synchronized with the pumping rhythm of the heart of the recipient. See Block **105**.

[0039] The controller **72** in the portable pump assembly **12** monitors the flow of blood and automatically stops the flow once 450 milliliters of blood have been pumped. See Block **107** and Block **108**. At the rate of 100 ml/min, the desired donation of 450 milliliters can be collected in only 4.5 minutes. If the flow of blood is pulsed in synchronization with the patient's heartbeat, the 450 milliliter of blood/plasma can be transferred in as little as 4 minutes. This is at least three times faster than a traditional gravity-fed blood transfusion from a blood bag. This increase in blood transfer efficiency greatly reduces the chances that a person who has lost blood will enter hemorrhagic shock or die of exsanguination. Since the volume of blood is monitored and controlled, the donor will not be drained of a dangerous amount of blood. The controller **72** will also monitor the flow rate and can sound alarms if the flow rate is too fast, indicating a disconnection or untreated hemorrhage, or too slow indicating a blockage. See Block **110** and Block **112**.

[0040] As a donor approaches his/her maximum donation volume, the controller **72** can display and/or sound a warning. This enables a medic to connect a different donor to the transfer tube **14** so that the initial donor can be safely disconnected. By connecting a second donor or bag to the transfer tube **14** in this manner, no air gets introduced into the transfer tube **14** and no time is wasted repriming the transfer tube **14**.

[0041] The controller **72** runs the operational software **66**. Referring to FIG. **7** in conjunction with FIG. **4** and FIG. **3**, the inputs of an exemplary version of the operational software **66** are shown. As is indicated by Block **120**, the emergency blood transfusion system **10** is powered on. Upon power up, the operation software **66** runs a rapid diagnosis that checks the status of the system and the power available in the batteries **65**. See Block **122** and Block **124**. If the system diagnosis is successful and there is power, a green light is displayed on the display. See Block **126**. This may also be accompanied with some sort of an audible signal.

[0042] The medical professional using the emergency blood transfusion system **10** is then prompted with a patient input prompt. In this prompt, the approximate weight of the patient is entered. See Block **128**. This can be done using a menu selection presented on the display **74**. For example, the prompt may indicate that the patient is 1. 100lbs-140 lbs, 2. 141 lbs-180 lbs, 3. 181 lbs-220 lbs 4. Over 220 lbs. The size of a patient is proportional to the volume of blood in that patent and can be used to determine a maximum transfusion volume.

[0043] The next prompt is a prompt to indicate if the donor source will be a live donor or a prefilled bag. See Block **130**. Once the donor source is selected, the user is prompted to connect the transfer tube **14** to a donor source and then insert the transfer tube **14** into the groove **56** on the housing **42**. See Block **132**. If the donor source is a person, the operational software **60** sets a limit of **450** milliliters of blood to be drawn. If the donor source is a bag, a second prompt is created that asks for the volume of the bag. See Block **134**.

[0044] Once the transfer tube **14** is set, the switch **69** in the flange slot **64** is triggered by the tube anchor **60**. This informs the controller **72** that the transfer tube **14** is properly positioned in the containment housing **42**. Once the transfer tube **14** is set, the operational software **66** generates a prompt asking if the transfer tube is connected to the donor source. See Block **136**. If the prompt is answered in the affirmative, the operational software **66** prompts the user to open the venting valve **34** and the electromechanical pump **44** runs for a few seconds to prime the transfer tube **14**. See Block **138**. Once the transfer tube **14** is primed, the venting valve **34** is closed. An over pressure is soon created in the transfer tube **14** that is detected by the pressure sensor **88** and the controller **72** stops the electromechanical pump **44**.

[0045] The controller **72** generates a prompt to attach the transfer tube **14** to the patient in need of blood. See Block **140**. Once the transfer tube **14** is connected a "start" prompt is answered on the display and the electromechanical pump **44** starts pumping. See Block **142**. If the controller **72** receives biofeedback from the patient, the pumping is synchronized to the heartbeat. See Block **144**. This is a dynamically updated process since the heart may start beating faster or slower as blood supply in the patient increases. The pressure in the transfer tune is monitored during pumping. See Block **146** in FIG. **7** and Blocks **110** and **112** in FIG. **6**.

[0046] Referring now to FIG. **8**, an alternate embodiment of the blood transfusion system **150** is shown. This blood transfusion system **150** is simplified and it can be carried in a sealed package **152** by a medic in the field. The blood transfusion system **150** includes a transfer tube **154**. An in-line filter **156** and a one way valve **158** are provided on the transfer tube **154**. The filter **156** and one way valve **158** prevents any clots or coagulations from passing into the patient and prevents any backflow into the donor.

[0047] The transfer tube **154** has a first end **160** and an opposite second end **162**. Both ends **160**, **162** terminate with connectors **164** that directly receive needle heads **166**. In this manner, the patient and the donor need not be prepared with

IV ports. Rather, the transfer tube **154** can be directly connected to a vein of the patient and an artery of the donor. Strips of tape **170** can be provided to hold the needle heads **166** in place and t**50** identify blood type.

[0048] The transfer tube **154** also has a tube anchor **172** formed along its length. The transfer tube **154** is engaged with a portable pump assembly **174**. The portable pump assembly **174** has a simplified design where the groove **176** for the transfer tube **154** is accessible on the top of the housing **178**. Thus, no opening and closing of the housing **178** is required. Otherwise, the portable pump assembly **150** operates in the same manner as has been previously described.

[0049] It will be understood that the embodiments of the present invention that are illustrated and described are merely exemplary and that a person skilled in the art can make many variations to those embodiments. All such embodiments are intended to be included within the scope of the present invention as defined by the below claims.

What is claimed is:

1. A method of rapidly transfusing blood directly from a donor to a patient, said method comprising:

providing a transfer tube;

intravenously connecting said transfer tube to the donor;

intravenously connecting said transfer tube to the patient;

providing a pump assembly that contains a pump, at least one battery for powering said pump, and a controller for controlling said pump, wherein said pump, said controller and said at least one battery are encased in a common housing;

engaging said transfer tube with said pump assembly wherein said pump, when activated, acts upon said transfer tube to move blood from the donor to the patient,

wherein said controller monitors blood volume moved by said pump and automatically stops said pump once a predetermined volume of blood has been transferred.

2. The method according to claim **1**, wherein said predetermined volume is between 400 milliliters and 450 milliliters

3. The method according to claim **1**, wherein said transfer tube has a first end, a second end and an uninterrupted length between said first end and said second end.

4. The method according to claim **3**, further including attaching an intravenous port to the patient, wherein said first end of said transfer tube terminates with a connector and wherein intravenously connecting said transfer tube to the patient includes connecting the connector to the intravenous port.

5. The method according to claim **3**, further including attaching an intravenous port to the donor, wherein said second end of said transfer tube terminates with a connector and wherein intravenously connecting said transfer tube to the donor includes connecting the connector to the intravenous port.

6. The method according to claim **3**, wherein said first end of said transfer tube terminates with a first connector and wherein intravenously connecting said transfer tube to the patient includes connecting the first connector to a first needle head, wherein said first needle head is used to produce a first intravenous blood connection with the patient.

7. The method according to claim **6**, wherein said second end of said transfer tube terminates with a second connector and wherein intravenously connecting said transfer tube to the donor includes connecting the second connector to a second needle head wherein said second needle head is used to produce a second intravenous blood connection with the donor.

8. The method according to claim **3**, wherein said transfer tube includes a vent valve proximate said first end and said method includes venting air from said transfer tube to remove air from the transfer tube after said transfer tube is intravenously connected to the donor.

9. The method according to claim **3**, wherein said controller monitors pressure in said transfer tube between said pump and said patient, and said controller sounds an alarm should said blood pressure fall outside a preselected range.

10. The method according to claim **1**, wherein said transfer tube has an anchor flange extending therefrom at a point between said first end and said second end, wherein said common housing includes a slot for receiving said anchor flange and a sensor that detects said anchor flange in said slot.

11. The method according to claim **1**, further including detecting a heart rhythm of the patient and utilizing said heart rhythm to control said pump.

12. The method according to claim **1**, further including providing a manual crank that is detachable from said common housing and utilizing said manual crank to power said pump should said at least one battery run out of power.

13. A method of rapidly transfusing fluid directly from a donor source to a patient, said method comprising:

providing a transfer tube having a first end, a second end, and an uninterrupted length between said first end and said second end;

connecting said transfer tube to the donor source;

intravenously connecting said transfer tube to the patient;

providing a pump assembly that contains a pump, at least one battery for powering said pump, and a controller for controlling said pump, wherein said pump, said controller and said at least one battery are encased in a portable waterproof housing;

engaging said transfer tube with said pump assembly wherein said pump, when activated, acts upon said transfer tube to move blood from the donor source to the patient,

wherein said controller monitors blood volume moved by said pump and automatically stops said pump once a predetermined volume of blood has been transferred.

14. The method according to claim **13**, wherein said predetermined volume is selectively programmed into said controller.

15. The method according to claim **13**, wherein said donor source is selected from a group comprising human donors, prefilled bags of blood, and prefilled bags of blood plasma

16. The method according to claim **13**, wherein said transfer tube includes a vent valve proximate said first end and said method includes venting air from said transfer tube to remove air from said transfer tube after said transfer tube is intravenously connected to the donor.

17. The method according to claim **13**, wherein said controller monitors pressure in said transfer tube between said pump and said patient, and said controller sounds an alarm should said pressure fall outside a preselected range.

18. The method according to claim **13**, wherein said transfer tube has an anchor flange extending therefrom at a point between said first end and said second end, wherein said housing includes a slot for receiving said anchor flange and a sensor that detects said anchor flange in said slot.

19. The method according to claim **13**, further including detecting a heart rhythm of the patient and utilizing said heart rhythm to control said pump.

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